



EPIS Topic Gap Analyzer

Prototype explainer and output guide

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1. What I built

I built a first working prototype of an EPIS Topic Gap Analyzer. The tool collects EPIS publication PDFs, extracts the text, scores the content across predefined topic categories, filters candidate gaps through a working-group relevance matrix, and generates both CSV outputs and a readable HTML report.

I designed the prototype as a decision-support tool. It does not replace editorial judgment. It helps me detect possible under-covered themes and turn them into article ideas that can be manually reviewed before being proposed to EPIS.

55

PDFs indexed

8

Working groups

11

Topic categories

18

Relevant gaps

2. Pipeline structure



The pipeline is modular. Each step creates an output file that can be inspected before the next step runs. This makes the project easier to debug, explain, and improve.

Step	Script	Purpose	Main output
1	main.py	Crawls EPIS working-group pages and downloads EPIS-hosted PDFs.	manifest.csv
2	build_publication_index.py	Turns the manifest into a structured publication index.	publication_index.csv
3	extract_pdf_text.py	Extracts readable text from each downloaded PDF.	text_index.csv and /texts
4	analyze_topic_categories.py	Counts topic-category terms in each document and group.	group_topic_scores.csv
5	generate_relevant_topic_gaps.py	Keeps only weak or absent topics that are relevant for the group.	relevant_topic_gaps.csv
6	generate_html_report.py	Creates a readable HTML report from the final CSV.	reports/epis_topic_gap_report.html

3. Main measures in the CSV report

The final CSV file is not meant to be read as raw statistics only. Each measure supports a specific interpretation. I use the scores to identify candidate gaps, then I review them manually before treating them as useful article ideas.

Column	Simple meaning	How I use it
group_label	The EPIS working group, for example South America, Europe, or Indo-Pacific.	It tells me where the possible gap belongs.
topic_label	The broad theme being evaluated, such as Energy security, China's role, or Technology/cyber/AI.	It identifies the area that may be under-covered.
coverage_level	A classification based on the group score compared with the corpus average. "Gap" means zero detected coverage; "weak" means clearly below average.	It gives a quick label for how under-covered the topic appears.
priority	A simplified ranking: High, Medium, or Low.	It helps me decide which candidate gaps deserve attention first.
group_score_per_1000_words	How often the topic appears in that working group, normalized per 1,000 words.	It avoids overvaluing groups that simply have more PDFs or more text.
corpus_average_per_1000_words	The average topic frequency across the whole EPIS corpus.	It provides the benchmark used for comparison.
gap_priority_score	The distance between the corpus average and the group score, when the group is below average.	Higher values suggest a stronger candidate gap.
document_count	The number of EPIS PDFs available for that working group in the corpus.	It gives context: a group with few documents needs extra caution.
interpretation	A plain-language explanation of the result.	It makes the CSV easier to read and easier to present.
possible_article_idea	A suggested article angle based on the detected candidate gap.	It turns the analysis into something useful for EPIS Basics planning.

4. Methodological decisions

I made three important methodological choices to keep the prototype understandable and defensible.

Decision	Reason
I used topic dictionaries instead of a black-box model.	This makes the first version transparent. I can explain exactly which terms define each topic category, and I can improve the dictionaries later.
I normalized counts per 1,000 words.	Groups do not have the same number of PDFs or the same amount of text. Normalization makes comparisons more reasonable.
I added a relevance matrix by group.	A low score is not automatically a gap. For example, migration may not be a central Arctic and Space topic. The relevance matrix prevents many false positives.
I kept human review as part of the workflow.	The tool identifies candidate gaps. A researcher still needs to decide whether the suggestion is intellectually useful and suitable for EPIS.

5. How I interpret the results

I read the results as a shortlist of possible article directions, not as final conclusions. A high-priority candidate means that a topic is both relevant to the working group and weakly represented in the current PDF corpus according to the dictionary-based scoring.

For example, if South America shows a weak score for migration or China's role, I do not treat that as automatic proof that EPIS has ignored the topic. I treat it as a prompt for manual review: I check the titles, compare the group mandate, and decide whether a new 500-word contribution would add value.

Important limitation

The prototype depends on the quality of the keyword dictionaries and on the available EPIS-hosted PDFs. Some EPIS publications may exist as web pages without PDFs, and those are not fully represented in the current text corpus.

6. Final deliverables produced by the prototype

Deliverable	Use
epis_topic_gap_report.csv	Structured report with candidate gaps, scores, interpretations, and article ideas.
reports/epis_topic_gap_report.html	Readable report suitable for internal demonstration. This is easier to present than a crowded CSV.
EPIS_Topic_Gap_Analyzer_Explainer.pdf	Short documentation note explaining what I built and how to interpret the measures.